



**ITM SKILLS
UNIVERSITY**

School of Future Tech

Case Study Report

on

LinguaHub

A Full-Stack Language Learning Platform

(Node.js · Express · MongoDB · Socket.io · Firebase · Groq AI)

by

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Source code: github.com/Yashhh710/linguahub

Live demo: linguahub-xi.vercel.app

Index

1. Introduction to the Case Study
2. Problem Statement / Case Background (Abstract)
3. Case Study Design
4. Methods, Algorithms & Technology Applied
5. Implementation Details and Snapshots
6. Results and Conclusion
7. References

1. Introduction to the Case Study

Learning a new language takes regular practice, and most learners drop out because practice material, progress tracking and motivation are spread across different tools. Apps that combine short lessons, quizzes, speaking practice and game-like rewards such as experience points (XP), streaks and badges help learners keep a daily habit.

This case study presents **LinguaHub**, a full-stack language-learning web platform. Learners register, follow structured lessons, take server-graded quizzes, practise speaking, earn XP and daily streaks, unlock achievements, add friends and compete on leaderboards.

Teachers can author lessons and quizzes, optionally drafting them with AI, and administrators manage users and view platform statistics.

The system is built with a Node.js and Express REST API, MongoDB through Mongoose, Socket.io for realtime updates, Firebase for Google sign-in and push notifications, and the Groq API for AI-assisted content authoring. The frontend is a plain HTML, CSS and JavaScript single-page application with no build step. It is deployed on Vercel at linguahub-xi.vercel.app.

2. Problem Statement / Case Background (Abstract)

Background

Many language apps show placeholder progress or fake demo data, trust scores sent by the browser, or compute streaks in the server's timezone, so a learner who practises late at night can lose a streak unfairly. A trustworthy learning platform needs real, server-verified progress data and fair day boundaries for every learner.

Problem Statement

Design and build a web platform where learners can study structured language lessons, test themselves with quizzes, practise speaking, and stay motivated through XP, levels, streaks, achievements, friends and leaderboards. Teachers must be able to create content, and administrators must be able to manage users. Every number shown to a learner must come from real logged events.

Objectives

- Provide secure registration and login (email/password and Google sign-in) with student, teacher and admin roles.
- Deliver structured, level-tagged (CEFR A1–C2) lessons and multiple-choice quizzes.
- Grade quizzes and speaking practice on the server, never trusting client-reported scores.
- Track XP, levels, daily streaks (in the learner's own timezone) and achievements.
- Support friends, friend requests and weekly / all-time leaderboards.
- Push XP, streak and friend events to open browser tabs in realtime.
- Let teachers author lessons and quizzes, with optional AI-generated drafts.

Abstract

LinguaHub is organised into a static single-page frontend and a layered Express API (middleware, routes, services, utilities and Mongoose models) backed by MongoDB. A central `awardXp()` service is the only place XP is granted: it updates the learner's total, logs an XP event, advances the daily streak, unlocks achievements and emits a Socket.io event. Quiz grading, speaking-accuracy scoring (normalised Levenshtein similarity) and level maths run on the server, and the pure logic is covered by 16 unit tests. The platform ships with 107 lessons across 97 languages (full content for Spanish, French and Japanese), 100 quizzes and 12 achievements, and exposes 11 REST route groups plus a realtime channel.

3. Case Study Design

The design separates the browser UI, the REST API and the data layer so each part can change independently.

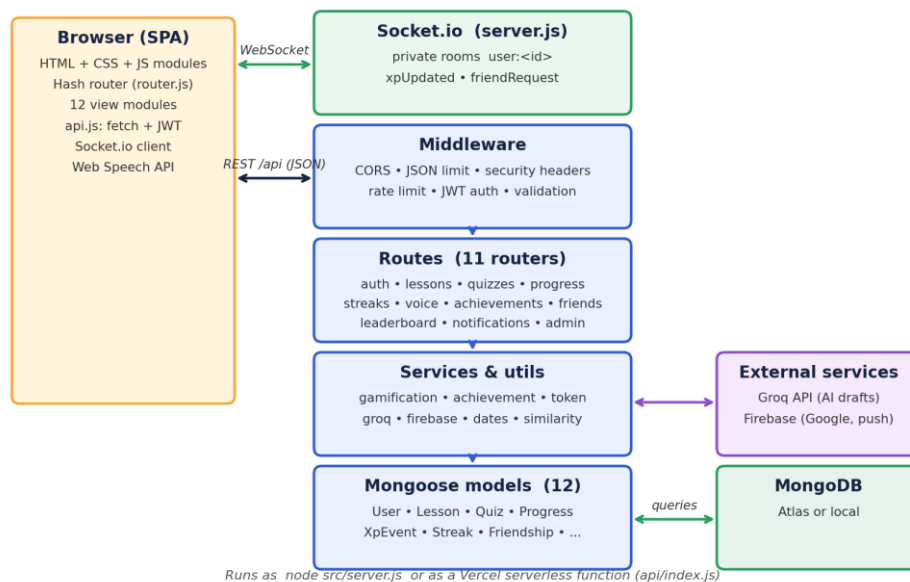


Figure 1: System architecture of LinguaHub

3.1 Project Structure

Folder / File	Responsibility
public/	Static frontend: index.html, css/ (5 stylesheets), js/ (router, state, api, shell, toast, skeleton, icons) and js/views/ (12 page modules).
src/app.js	Builds the Express app: CORS, JSON parsing (200 kb limit), security headers, rate limits, the 11 routers, static hosting and the error handler.
src/server.js	Starts the HTTP server, attaches Socket.io, connects MongoDB, handles graceful shutdown.
src/config/	env.js (environment loading and validation) and db.js (MongoDB connection).
src/models/	12 Mongoose schemas (User, Lesson, Quiz, QuizAttempt, Progress, XpEvent, Streak, and others).
src/routes/	One router per feature area; handlers live directly in the route files.

Folder / File	Responsibility
src/services/	gamificationService, achievementService, tokenService, groqService, firebaseService.
src/middleware/	auth (JWT and roles), validate, rateLimit, errorHandler, notFound.
src/utils/	Pure, unit-tested logic: dates, gamification maths, text similarity, plus asyncHandler and httpError.
data/curriculum/	Real seed content as JSON: lessons, quizzes, achievement catalogue and language list.
api/index.js, vercel.json	Serverless entry point and rewrite rule used for the Vercel deployment.
tests/, scripts/, docs/	Unit tests, curriculum import and make-admin scripts, API reference.

3.2 Data Model

All user-owned collections reference the User document by id. The XpEvent collection is a ledger of every XP award and drives the weekly chart and weekly leaderboard.

Collection	Key fields	Purpose
User	name, email, password (bcrypt), role, authProvider, learningLanguages, dailyGoalMinutes, timezone, xp, reminder	Account, role and XP total
Lesson	title, language, level (A1–C2), topic, xpReward, order, content{vocabulary, grammar, examples, tip}, source	Lesson content
Quiz	title, language, level, xpReward, questions[4 options + correctAnswer + explanation], source	Reusable quizzes
QuizAttempt	user, quiz, score, total, percent, passed, xpEarned, answers	Graded attempt history
Progress	user, lesson, completed, timeSpentSec, xpAwarded (unique per user+lesson)	Per-lesson completion
XpEvent	user, xp, source (lesson / quiz / voice), ref, label, language	XP ledger
Streak	currentStreak, longestStreak, freezesAvailable, lastActiveDay, days[], protectedDays[]	Daily streak state
VoicePractice	user, language, expectedText, heardText, accuracy, xpEarned	Speaking attempts
Achievement / UserAchievement	key, title, category, goal / user + achievement (unique)	Badge catalogue and unlocks
Friendship	requester, recipient, status (pending / accepted / declined / blocked)	Friend graph
Notification	user, message, read	In-app notifications

3.3 REST API Design

All responses use the envelope { success, data, message }. Protected routes expect an Authorization: Bearer token. The full reference is in docs/API.md.

Route group	Main endpoints	Access
/api/auth	POST register, login, google; GET/PATCH /me; PUT /me/password, /me/reminder	Public / signed-in
/api/lessons	GET list (language, level filters), GET /:id, POST, PUT, DELETE, POST /generate (AI)	Read public; write teacher/admin
/api/quizzes	GET list, GET /:id (answers hidden), POST /generate, POST, POST /:id/attempts	Attempts need sign-in
/api/progress	GET summary (totals, weekly XP, recent); PUT /lessons/:id to complete	Signed-in
/api/streaks	GET current, longest, freezes and 30-day calendar	Signed-in
/api/voice	POST /attempts, GET /attempts/me	Signed-in
/api/achievements	GET catalogue, GET /me with unlocked state	Public / signed-in
/api/friends	GET list, GET /search, POST /requests, PUT /requests/:id, DELETE /:id	Signed-in
/api/leaderboard	GET ?scope=global friends &period=week alltime	Public / signed-in
/api/notifications	GET, PUT read, POST /announce (broadcast)	Signed-in; announce teacher/admin
/api/admin	GET /stats, GET /users, PUT /users/:id/role	Admin only

3.4 Frontend Design

The frontend is a single-page app that uses ES modules and a hash router (router.js). Unauthenticated visitors are redirected to the login page, and signed-in users are redirected away from it. A shared shell renders the sidebar, the streak and XP chips and the notification bell. Views are separate modules: Dashboard, Lessons, Lesson detail, Practice (quizzes), Quiz, Progress, Friends, Achievements, Profile, Studio (teachers) and Admin. The Studio and Admin links appear only for the matching roles.

3.5 Security and Reliability Design

- Passwords are hashed with bcrypt; sessions use JWT (7-day default expiry); role checks use requireRole().
- Input is validated with express-validator; mongoose schemas enforce limits such as four options per question and 1–30 questions per quiz.
- Basic hardening: X-Content-Type-Options, X-Frame-Options and Referrer-Policy headers, a 200 kb JSON limit and configurable CORS origins.
- In-memory rate limiting: 10 requests per minute for login and register, 6 per minute for AI generation.
- The server refuses to start, with a readable message, if MONGODB_URI or JWT_SECRET is missing or malformed.

4. Methods, Algorithms & Technology Applied

4.1 Key Methods and Algorithms

Level calculation. Level n begins at $100 \times (n - 1)^2$ XP, so levels start at 0, 100, 400, 900 and 1600 XP. The level is $\text{floor}(\sqrt{(xp / 100)}) + 1$.

Daily streak with freezes. Day boundaries are computed in the learner's own IANA timezone using `Intl.DateTimeFormat`. Activity on consecutive days extends the streak; missing exactly one day is forgiven if the learner has a streak freeze (the missed day is stored as a protected day); any longer gap resets the streak to 1. A freeze is earned each time the streak passes a multiple of 7 days, with a maximum of one held. The streak shown to the learner drops to 0 automatically if they skipped yesterday and have not yet practised.

Server-side quiz grading. The API hides correct answers when a quiz is fetched. On submission it compares answers with the stored key. A score of 70% or more passes and earns $\text{reward} \times \text{percent}$ XP; a lower score earns 20% of the reward. The review with explanations is returned only after grading.

Speaking practice scoring. The browser converts speech to text with the Web Speech API and posts the transcript. The server normalises both texts (lower-case, accents and punctuation removed) and computes $1 - \text{Levenshtein distance} / \text{max length}$ as a 0–100 accuracy. It also returns a per-word match report. 90% or more earns 10 XP and 70% or more earns 5 XP.

Achievements. After every XP award the server compares the learner's streak, XP, completed lessons, passed quizzes and friend count with the 12-badge catalogue (categories streak, xp, lessons, quiz, social) and inserts a `UserAchievement` for each newly reached goal. A unique index prevents duplicates under concurrent requests.

Leaderboards. The all-time board sorts users by total XP. The weekly board aggregates the last 7 days of `XpEvent` records. Both can be limited to the learner's accepted friends.

AI content generation with fallback. Teachers can draft lessons, and learners can generate practice quizzes, through the Groq chat-completions API with JSON-only output. The service tries a list of models in order (openai/gpt-oss-120b, llama-3.3-70b-versatile, openai/gpt-oss-20b) with a 20-second timeout per call and returns a clear error if all fail. Without a `GROQ_API_KEY` the endpoints return HTTP 503 and the rest of the app is unaffected.

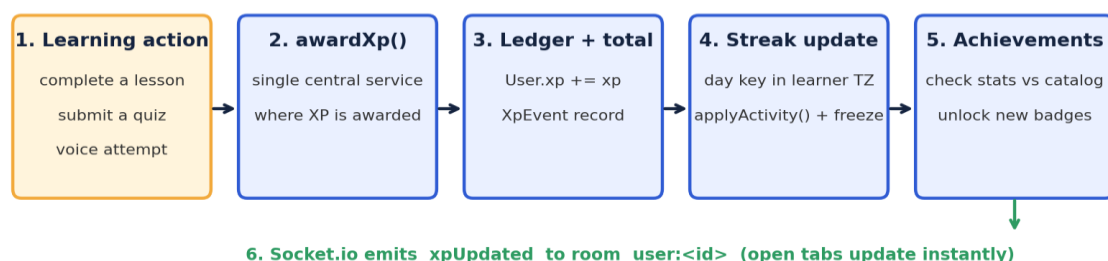


Figure 2: What happens on every XP award

4.2 Technology Stack

Category	Details
Runtime and language	Node.js 18 or later, JavaScript (CommonJS on the server, ES modules in the browser)
Web framework	Express 4.21 with cors and express-validator 7
Database	MongoDB (Atlas or local) with Mongoose 8
Authentication	JSON Web Tokens (jsonwebtoken 9), bcryptjs, Firebase Admin 12 for Google sign-in
Realtime	Socket.io 4.8 (private rooms per user)
Notifications	Firebase Cloud Messaging push plus in-app notifications
AI	Groq API (OpenAI-compatible chat completions)
Frontend	HTML5, CSS3 (design tokens), vanilla JavaScript SPA, Web Speech API, no build step
Testing	Node built-in test runner (node --test)
Hosting and tools	Vercel (serverless), nodemon, dotenv, Git and GitHub

5. Implementation Details and Snapshots

5.1 Gamification Rules (pure functions)

The XP, level and streak rules live in `src/utils/gamification.js` as pure functions with no database access, so they are easy to unit test.

```
const levelFromXp = xp => Math.floor(Math.sqrt(Math.max(0, xp) / 100)) + 1;
const xpForLevel = level => 100 * (level - 1) ** 2;

function applyActivity(state, today) {
  if (state.lastActiveDay === today) return { ...state, changed: false };
  let current, freezes = state.freezesAvailable || 0, protectedDay = null;
  if (state.lastActiveDay === shiftDay(today, -1)) {
    current = (state.currentStreak || 0) + 1;
  } else if (state.lastActiveDay === shiftDay(today, -2) && freezes > 0) {
    current = (state.currentStreak || 0) + 2; // missed day covered by a freeze
    freezes--; protectedDay = shiftDay(today, -1);
  } else {
    current = 1;
  }
  /* earn a freeze each time the streak crosses a multiple of 7 ... */
}
```

5.2 Central XP Service

`awardXp()` is the single place XP is granted. It is called from the lesson, quiz and voice routes.

```
async function awardXp(io, { userId, xp, source, ref, label, language }) {
  const user = await User.findById(userId);
  if (xp > 0) { user.xp += xp; await user.save();
    await XpEvent.create({ user: userId, xp, source, ref, label, language }); }
  const today = dayKey(new Date(), user.timezone); // learner-local day
  const result = applyActivity(streak, today); // advance the streak
  const unlocked = await checkAndUnlockAchievements(userId, { xp: user.xp, streak });
  if (io) io.to(`user:${userId}`).emit("xpUpdated", payload); // realtime push
}
```

```
    return payload;
  }
}
```

5.3 Server-Side Quiz Grading

```
const score = quiz.questions.reduce((t, q, i) => t + (answers[i] === q.correctAnswer ? 1 : 0), 0);
const percent = Math.round((score / quiz.questions.length) * 100);
const passed = percent >= 70;
const xpEarned = passed ? Math.round(quiz.xpReward * (percent / 100))
  : Math.round(quiz.xpReward * 0.2);
```

5.4 Speech Similarity Scoring

```
function score(expected, heard) {
  const a = normalize(expected).replace(/\s/g, ""), b = normalize(heard).replace(/\s/g, "");
  const max = Math.max(Array.from(a).length, Array.from(b).length);
  return Math.max(0, Math.round((1 - levenshtein(a, b) / max) * 100));
}
```

5.5 Setup, Seeding and Testing

The application is configured with a .env file (MONGODB_URI and JWT_SECRET are required; GROQ_API_KEY and FIREBASE_* are optional). The commands below install, seed, test and run it:

```
npm install
npm run import:curriculum # idempotent upsert: lessons, quizzes, achievements
npm run make-admin -- you@example.com
npm test # 16 unit tests, no database needed
npm start # node src/server.js (or: npm run dev)
```

Running npm test executes 16 tests for the pure logic, and all 16 passed. They cover timezone day keys, day shifting, level thresholds, streak extension, reset after a gap, same-day activity, freeze handling and speech-similarity scoring.

Seed content	Count	Notes
Languages listed	97	languages.json
Lessons imported	107	13 full lessons (Spanish 5, French 4, Japanese 4) plus one starter lesson for each of the other 94 languages
Quizzes	100	5 quizzes for each of 20 languages
Achievements	12	Streak, XP, lessons, quiz and social categories

5.6 Application Snapshots

The snapshots below were captured from LinguaHub running locally with the curriculum imported and a few demo accounts (students, a teacher and an admin) created through the app's own API. They show the real interface, with demo learner activity.

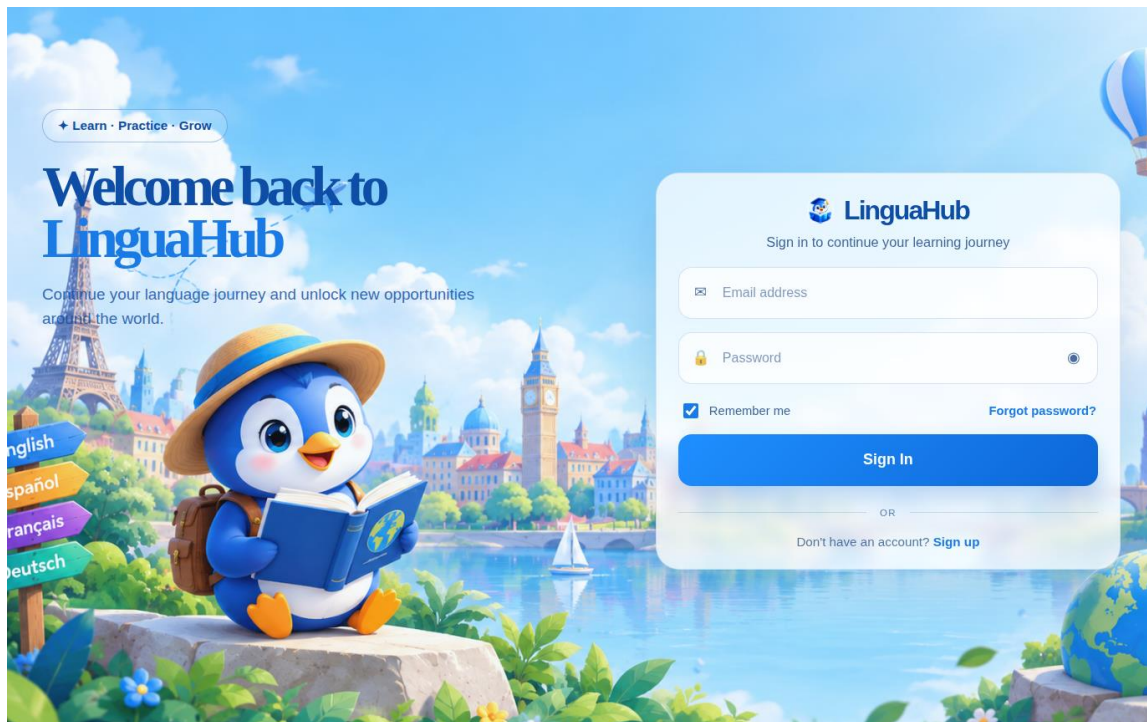


Figure 3: Login page

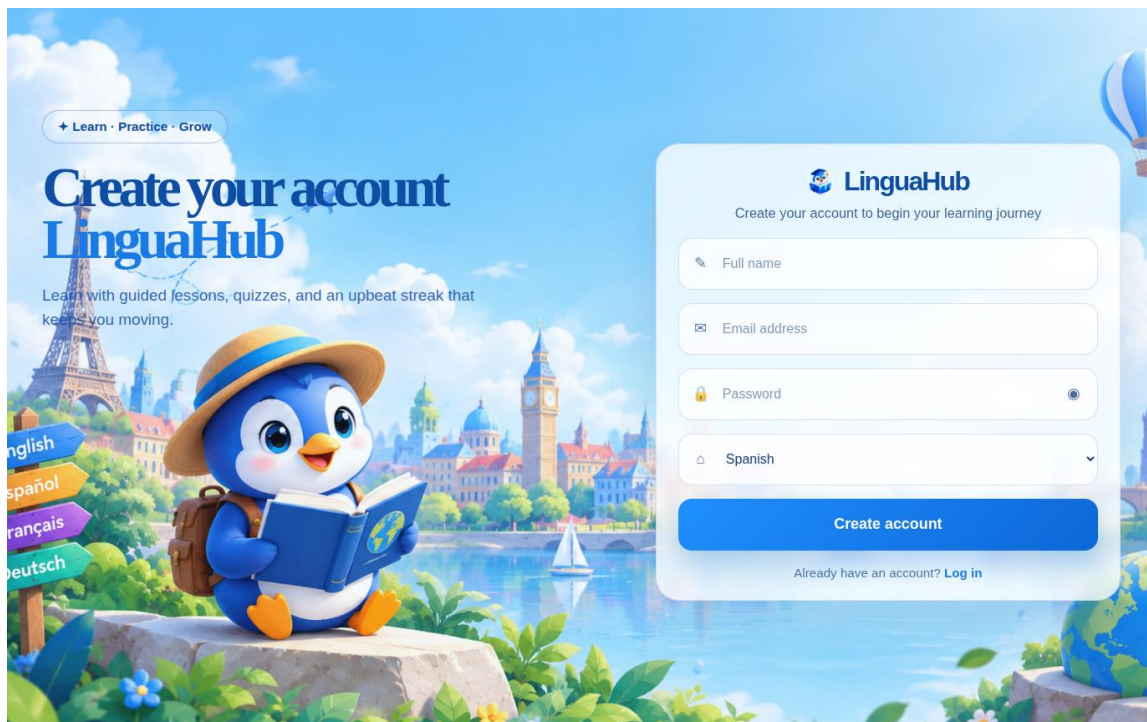


Figure 4: Registration page

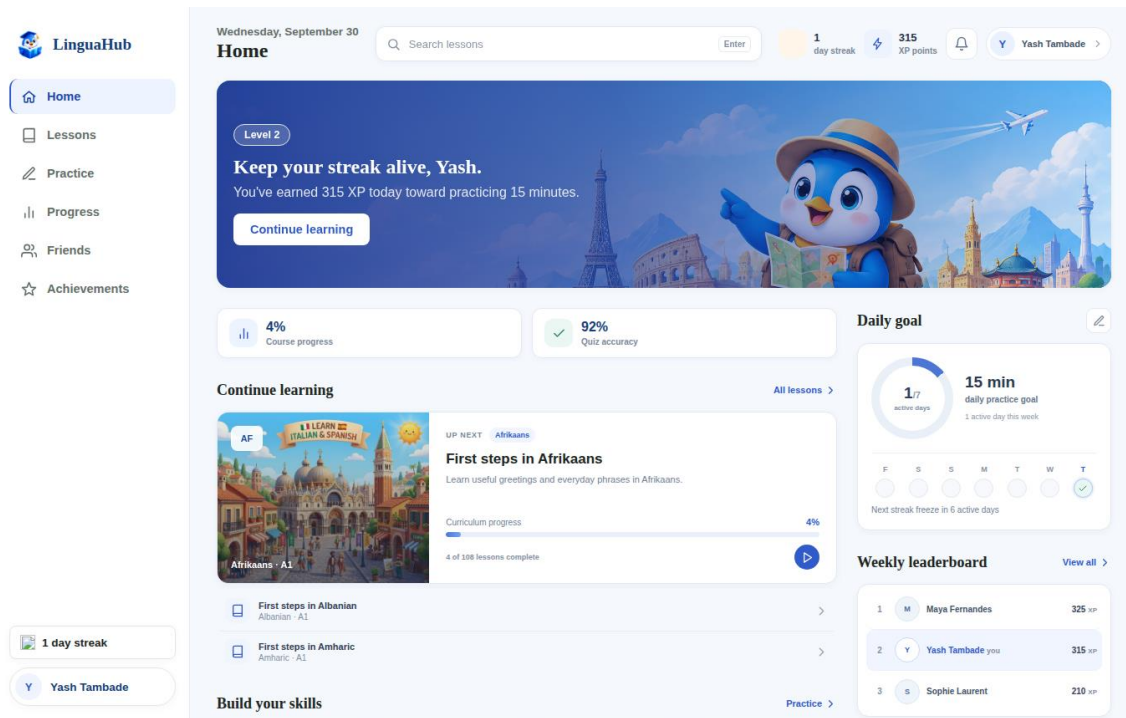


Figure 5: Dashboard with XP, streak, daily goal and weekly leaderboard

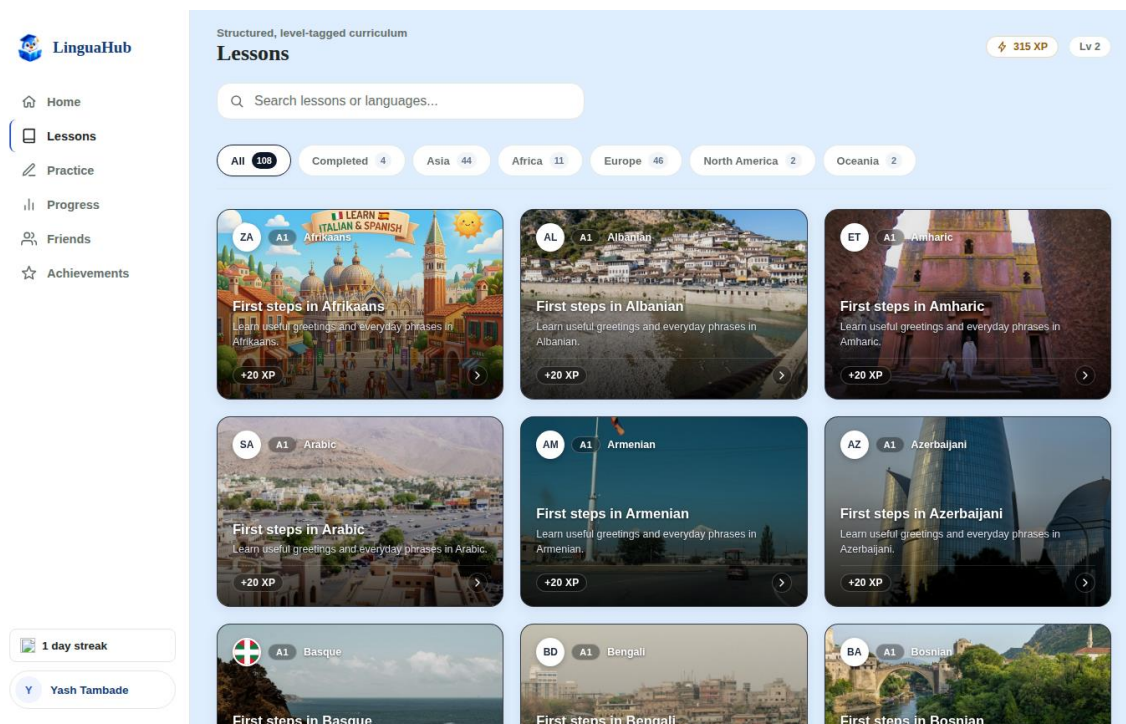


Figure 6: Lesson library filtered by region and language

- Home
- Lessons
- Practice
- Progress
- Friends
- Achievements

Vocabulary, grammar, and speaking practice

Lesson

... Back to lessons

A1 ES Spanish Completed

Greetings & Introductions

Learn useful phrases and everyday expressions in Spanish.

8 min 8 phrases +20 XP

Review Lesson

Lesson Content Vocabulary Grammar Examples Notes Resources

Vocabulary

Learn and practice essential words and phrases.

1 of 8

Hola

/OH-lah/

¡Hola! ¿Cómo estás?

Speaking Practice

Improve your pronunciation and confidence.

Practice "Hola"

Tap the mic and say the word out loud.

Lesson completed

Grammar

Learn the basic rules and structure of Spanish.

Subject pronouns with 'ser' (to be)

Spanish often drops the subject pronoun because the verb ending already shows who is speaking: (yo) soy, (tú) eres, (él/ella) es. Use 'soy' for permanent identity — 'Soy Ana' (I am Ana) — not 'estoy', which is for temporary states.

Examples

See how the words are used in real conversations.

Hola, me llamo Sofía. ¿Y tú?

Hi, my name is Sofia. And you?

Mucho gusto, Sofía. Soy Diego.

Nice to meet you, Sofia. I'm Diego.

1 day streak

Yash Tambade

Figure 7: Lesson detail with vocabulary, grammar, examples and speaking practice

- Home
- Lessons
- Practice
- Progress
- Friends
- Achievements

Search quizzes, languages or topics...

315 XP Lv 2

QUIZ ARENA - DAILY CHALLENGE

Daily Challenge

Keep your streak alive — one quiz a day builds real fluency.

1 day streak +40 XP reward 1 / 7 days this week

Start Daily Quiz QuizBlast

Your Progress

View all

Lv 2 Beginner 215 / 300 XP

1 Day Streak

Keep it going!

Next streak freeze in 6 active days

Top Learners

View all

1 Maya Fernandes 325 XP

2 Yash Tambade (you) 313 XP

3 Sophie Laurent 218 XP

Create AI Quiz

Generate a quiz on any topic in language.

All Categories Vocabulary Language Basics

All Languages Random order

IT A1

Greetings Check-in

Practice greetings with real feedback.

Italian

5 questions +40 XP

TH A2

Travel Practice

Practice travel with real feedback.

Thai

5 questions +45 XP

PT A2

Travel Practice

Practice travel with real feedback.

Portuguese

5 questions +45 XP

ES A2

Travel Practice

Practice travel with real feedback.

Spanish

5 questions +45 XP

Completed - 100%

DE B1

Daily Conversation

Practice daily life with real feedback.

German

5 questions +50 XP

FR A1

Greetings Check-in

Practice greetings with real feedback.

French

5 questions +40 XP

NL A2

Food Order Practice

Practice food with real feedback.

Dutch

5 questions +45 XP

SA A2

Food Order Practice

Practice food with real feedback.

Arabic

5 questions +45 XP

TH A2

Food Order Practice

Practice food with real feedback.

Thai

5 questions +45 XP

IN A2

Food Order Practice

Practice food with real feedback.

Hindi

5 questions +45 XP

TH A1

Numbers Practice

Practice numbers with real feedback.

Thai

5 questions +45 XP

IL A1

Numbers Practice

Practice numbers with real feedback.

Hebrew

5 questions +45 XP

1 day streak

Yash Tambade

Figure 8: Practice page with quiz library and daily challenge

Yash Tambade | 150096725173 | LinguaHub | Page 12

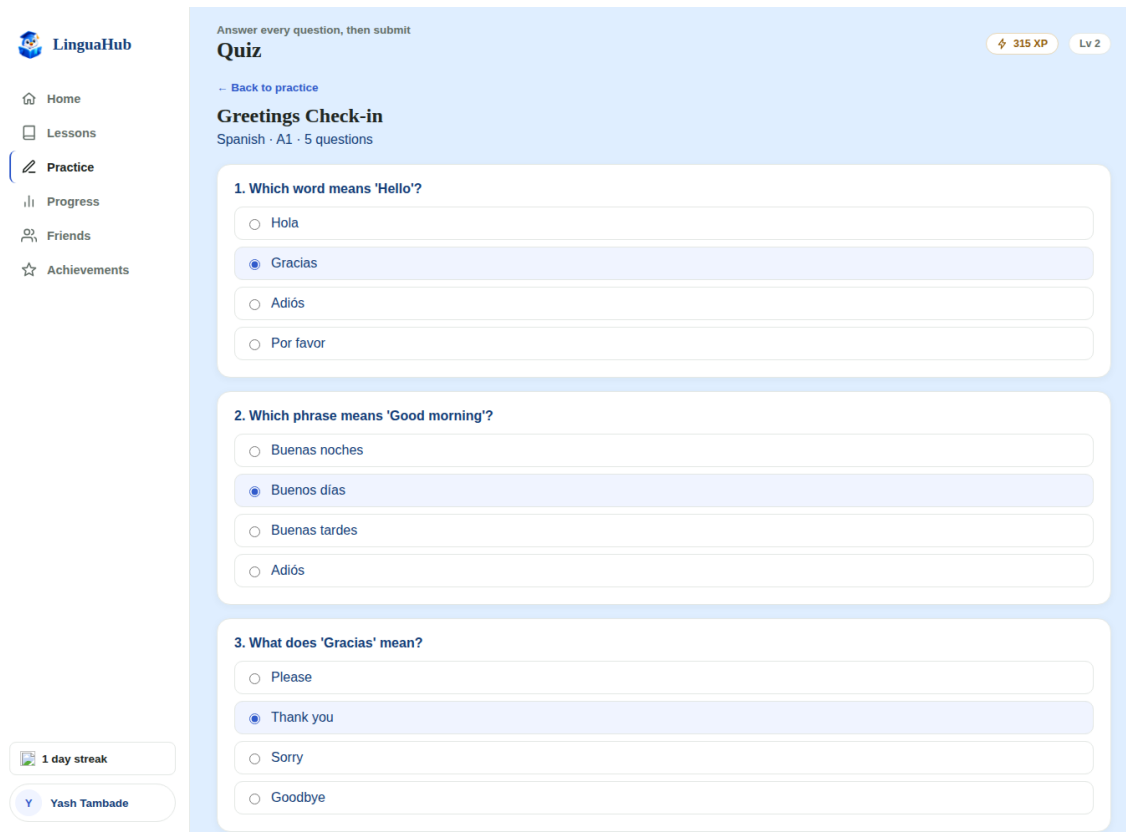


Figure 9: Taking a multiple-choice quiz

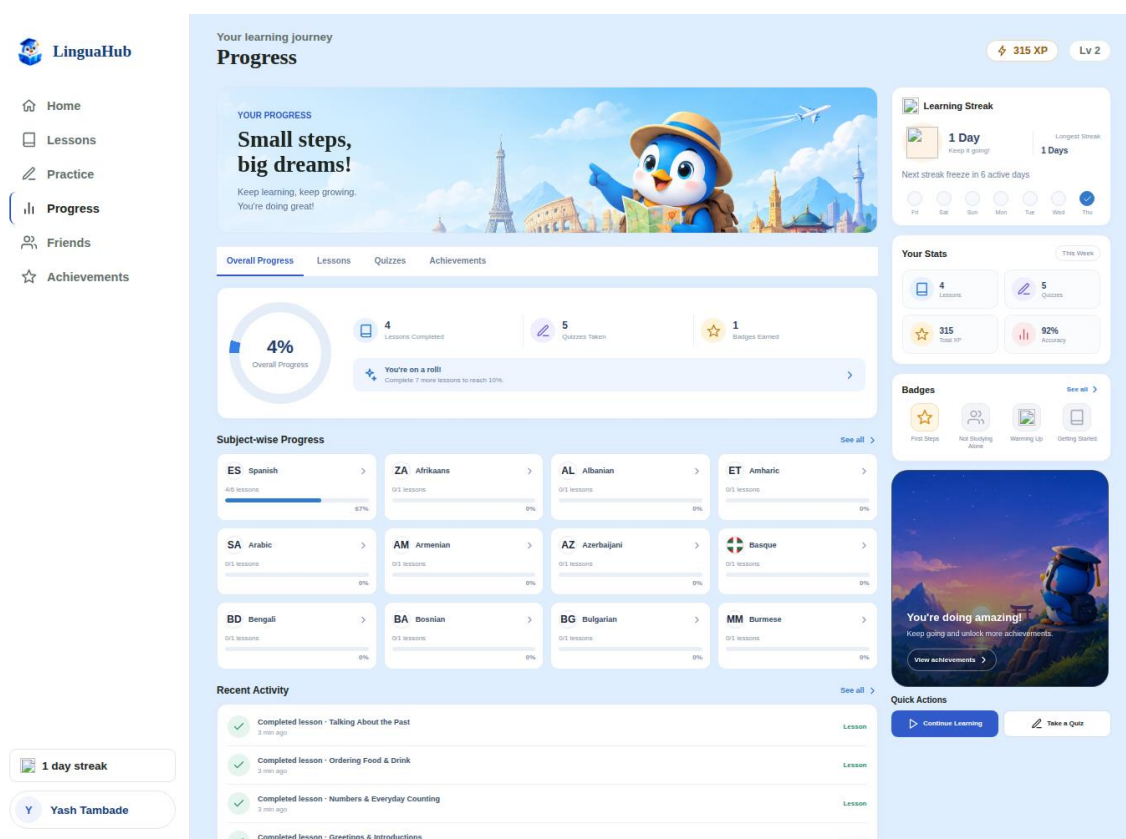


Figure 10: Progress page with weekly XP, skills and recent activity

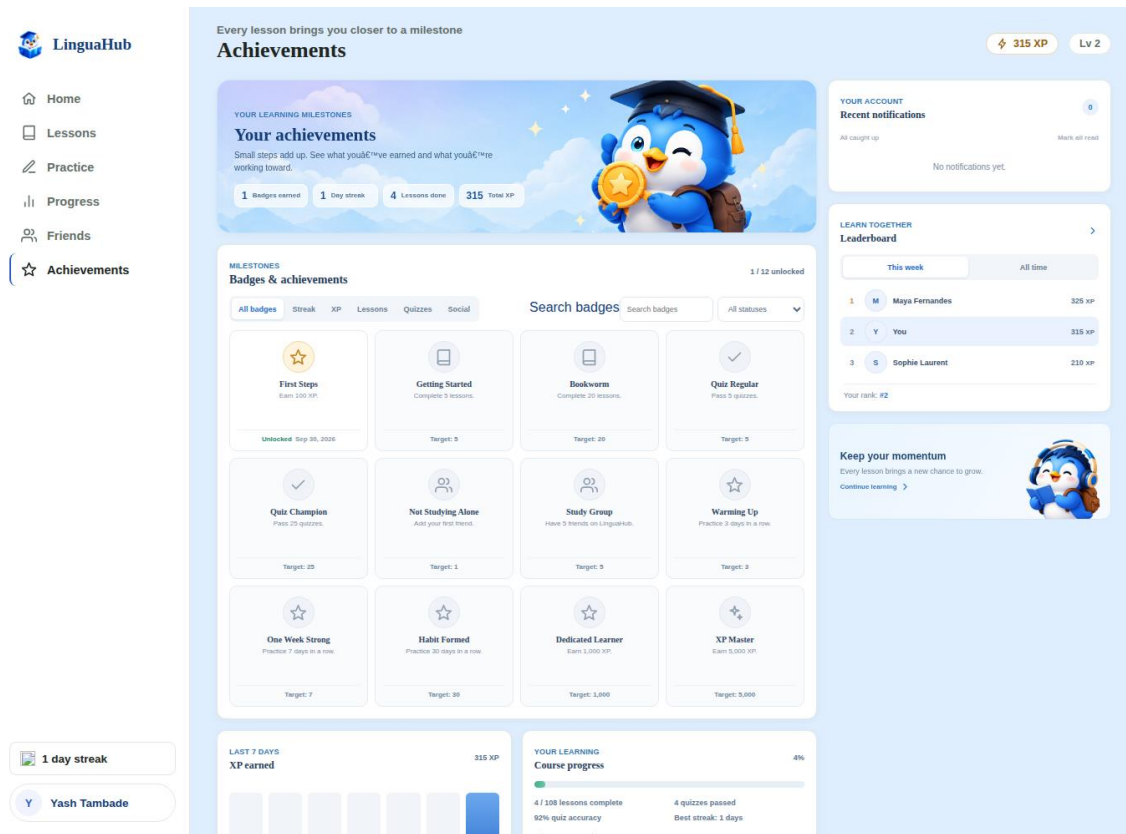


Figure 11: Achievements with badges and leaderboard

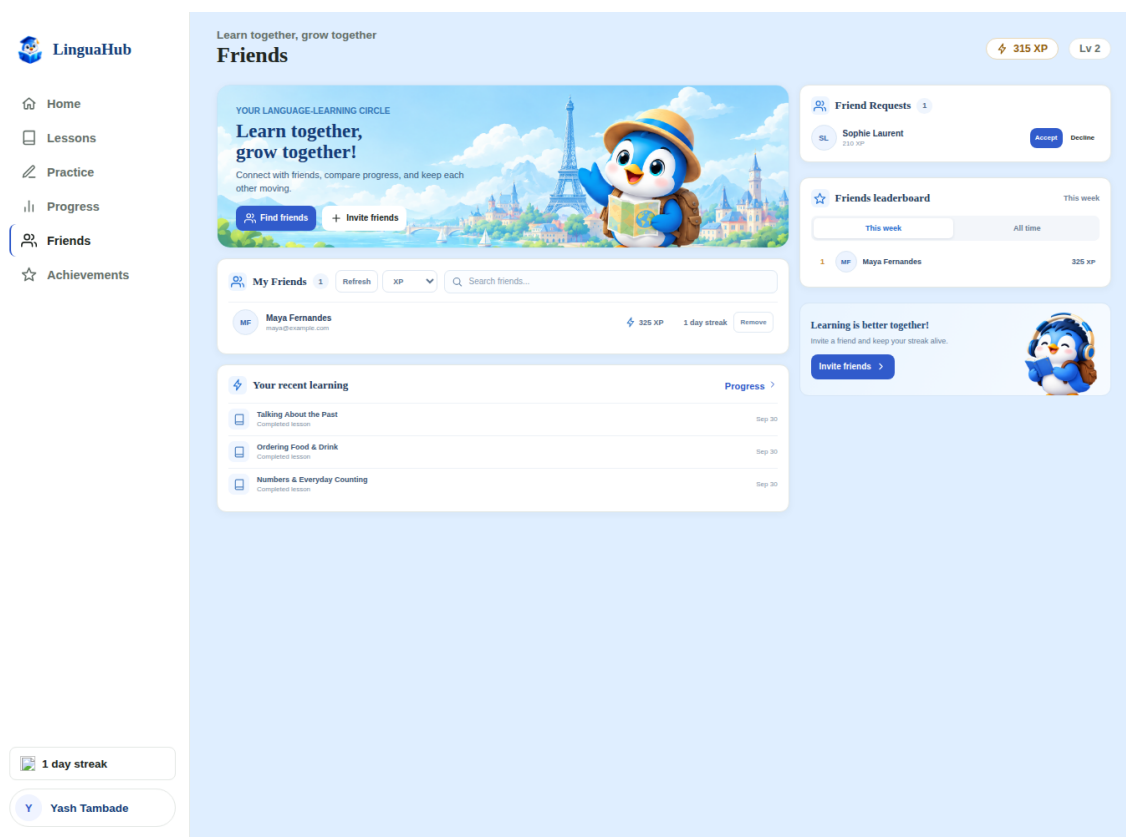


Figure 12: Friends, friend requests and friends leaderboard

Home

Lessons

Practice

Progress

Friends

Achievements

Y

Yash Tambade

Level 2 · 315 XP

Profile picture

+ From gallery

Choose a preset or add a photo.

Save photo

Account details

Name

Yash Tambade

Native language

English

Daily goal (minutes)

15

Save changes

Reminders

Daily practice reminder

A push notification if you haven't practiced yet.

☐ On

315 XP

Lv 2

Change password

Current password

New password

Update password

Statistics

Total XP

315

Level

2

Sign out

1 day streak

Yash Tambade

Figure 13: Profile, account settings and daily reminder

Home

Lessons

Practice

Progress

Friends

Achievements

Studio

Create lessons and quizzes for your students

0 XP

Lv 1

Studio

New lesson

New quiz

Draft with AI

Generate a starting point, then edit before publishing.

Language

Spanish

Level

A1

Topic

At the airport

Generate draft

Lesson

Title

Description

Language

Level

A1

Vocabulary (one per line: term | translation)

Hola | Hello

Grammar explanation

Study tip

Publish lesson

0 day streak

Rohan Mehta

Figure 14: Teacher Studio with AI draft and lesson publishing

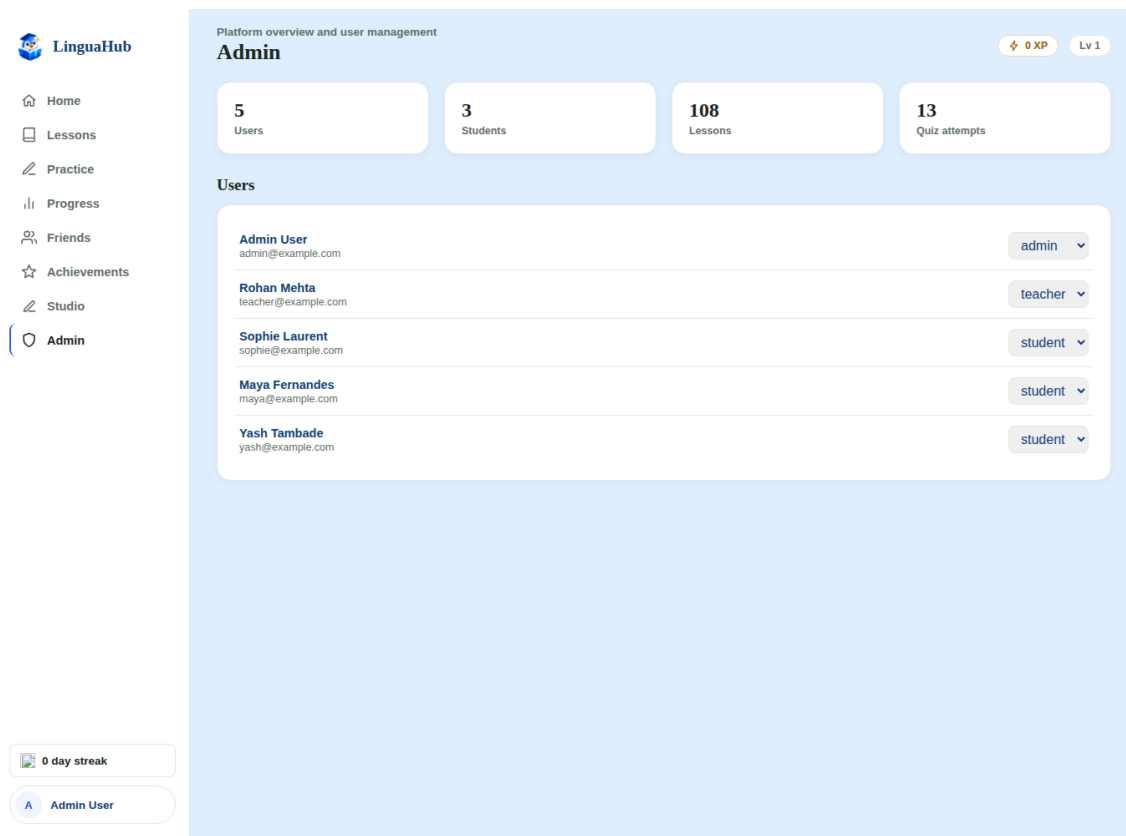


Figure 15: Admin dashboard with platform stats and role management

6. Results and Conclusion

Findings

- All planned features work end to end: authentication with roles, lessons, server-graded quizzes, speaking practice, XP and levels, timezone-aware streaks, achievements, friends, leaderboards, notifications, teacher authoring and an admin panel.
- Progress data is real. A new account starts at 0 XP and an empty streak, and every chart and leaderboard is computed from the XpEvent ledger.
- Trust-sensitive logic (quiz grading, speaking score, level and streak maths) runs on the server, and the pure parts are covered by 16 passing unit tests.
- The layered structure (middleware, routes, services, utils, models) kept the code small: about 1,800 lines for the backend, scripts and tests.
- The platform degrades gracefully: without a Groq key AI endpoints return a clear 503, and without Firebase config only Google sign-in and push are disabled.

Limitations

- Only Spanish, French and Japanese have full lessons. The other 94 languages have a single starter lesson whose translations are placeholders (for example "Hello in Afrikaans").
- Quiz achievements (Quiz Regular, Quiz Champion) cannot currently unlock, because awardXp() does not pass the passed-quiz count to the achievement check. Social

badges are re-checked only at the next XP award, not when a friend request is accepted.

- Socket.io needs a long-running server. The Vercel serverless entry point does not attach it, so realtime pushes work with `node src/server.js` but not on the serverless deployment.
- The rate limiter is in-memory per process, and quizzes and voice attempts can be repeated to earn XP again.
- Automated tests cover only the pure utilities; the API routes have no integration tests.

Future Scope

- Complete lesson content for more languages, and audio pronunciation for vocabulary.
- Fix the achievement triggers, add anti-farming limits for repeat quiz and voice XP, and a Redis-backed rate limiter.
- Integration tests for the API, and a persistent realtime host (or a hosted realtime service) for the Vercel deployment.
- Spaced-repetition review and adaptive quiz difficulty.

Conclusion

LinguaHub shows how a complete, honest learning platform can be built with a small, well-layered Node.js and MongoDB codebase. By awarding XP through one service, grading on the server, following each learner's own timezone and basing every statistic on logged events, it gives learners trustworthy feedback and motivation. It also gives teachers and administrators practical tools, and provides a solid base for richer content and scale.

7. References

- Node.js Documentation, nodejs.org/docs (including the built-in test runner).
- Express.js Documentation, expressjs.com.
- MongoDB Manual and Mongoose Documentation, mongodb.com/docs and mongoosejs.com.
- Socket.io Documentation, socket.io/docs.
- Firebase Admin SDK and Cloud Messaging Documentation, firebase.google.com/docs.
- Groq API Documentation, console.groq.com/docs.
- Vercel Documentation: Serverless Functions and Rewrites, vercel.com/docs.
- MDN Web Docs: Web Speech API and `Intl.DateTimeFormat`, developer.mozilla.org.
- M. Jones, J. Bradley, N. Sakimura, JSON Web Token (JWT), RFC 7519, IETF, 2015.
- V. I. Levenshtein, "Binary codes capable of correcting deletions, insertions and reversals", Soviet Physics Doklady, 1966.
- Council of Europe, Common European Framework of Reference for Languages (CEFR levels A1–C2).
- Project source code: github.com/Yashhh710/linguahub. Live demo: linguahub-xi.vercel.app.